

Urine Analyzer Lite

Project Status & Engineering Report — Levram Lifesciences

Device: automated 10-pad urine reagent-strip reader (reflectance photometry) · Regulatory pathway: India / CDSCO · Repo: github.com/Sid93/urine-analyzer-lite · Report date: 2026-05-21

Executive summary. The project is a benchtop/point-of-care urine dipstick analyzer ("Design A": ESP32-S3 + dual TCS34725 reflectance optics, motorized strip transport, thermal printout, SPI touch display). A complete firmware draft, India-sourced BOM, and a Blueprint mechanical concept already existed. Since then we have **backfilled the regulatory foundation (Stage 0)**, **integrated the research and design inputs (Stages 1 & 3)**, and **completed the electrical engineering groundwork (Stage 4)** — reconciling three conflicting design sources into one verified pin map, fixing real hardware bugs, locking the open design decisions, and generating an **ERC-clean schematic** and a **net-assigned starting PCB**, both validated with KiCad 10. The design is now coherent and ready for an electronics engineer to verify, route, and prototype.

1. The device (Design A)

Attribute	Specification
Function	Semi-quantitative reading of the standard 10-pad urine reagent strip (leukocytes, nitrite, urobilinogen, protein, pH, blood, specific gravity, ketones, bilirubin, glucose)
Principle	Diffuse reflectance photometry; dual TCS34725 RGB sensors + high-CRI LED; HSV classification
MCU	ESP32-S3 (USB-CDC), firmware state machine + sensor drivers complete (first draft)
Mechanics	GA12-N20 motor strip transport w/ encoder + limit switches; 3D-printed chassis/optical chamber
Display	3.5" SPI TFT (ILI9488) + XPT2046 touch decided
Output / extras	CSN-A2 58mm thermal printer; 275nm UV-C tray sterilization; LiPo + USB-C charging
Setting / class	Clinic / point-of-care; CDSCO best-guess Class B (to confirm)
Cost target	<\$250/unit at 50-unit volume (prototype ~₹387/unit, India-sourced)

2. Workflow & where we are

The project follows a 12-stage medical-device workflow with two hard gates: **Stage 0 gates everything; design freeze (Stage 5) gates all validation.**

Stage	Status
0 — Foundation (TPP, class, standards, risk, FTO)	DONE (backfilled)
1 — Literature & report (SciSpace)	DONE (integrated)

2 — Interactive explainer site	not started (optional)
3 — Device design iterations (Blueprint)	DONE (integrated)
4 — Engineering files (schematic, netlist, PCB)	DONE (this report) — pending engineer sign-off
5 — Prototype & bench verify → design freeze	NEXT
6–12 — Analytical validation → regulatory → mfg → post-market	future

3. Stage 0 — Regulatory foundation (regulatory/)

- **Target Product Profile** — intended use, analytes, performance targets, open issues.
- **CDSCO classification & pathway** — best-guess **Class B**; MD-12/13 test licence → MD-5 manufacturing licence route. **Confirm via regulatory consultant pre-submission before generating validation data.**
- **Standards matrix** — applies: ISO 13485, ISO 14971, IEC 61010-1/-2-101, IEC 61326-2-6, IEC 62304, IEC 62366-1, ISO 18113, CLSI EP05/06/07/17, IEC 62471 (UV-C). Dropped: ISO 10993, IEC 60601, ISO 15197.
- **ISO 14971 risk file** — 12-hazard FMEA seed; UV-C exposure (H1) is the highest-novelty hazard.
- **Freedom-to-operate** — preliminary optics-patent search. **Finding:** a white-LED + RGB-sensor reflectance head is textbook prior art, so "optics" is not yet a defensible differentiator; a specific novel optical scheme is needed to be patentable.

4. Stage 4 — Engineering (the core of recent work)

4.1 Source reconciliation

The electrical design lived in three places that **disagreed** (firmware `config.h`, Blueprint netlist, BOM): motor driver (MX1508 vs L298N), I2C pins, motor/HX711/printer GPIOs, display. We adopted `config.h` as the source of truth and produced one master pin map (`hardware/ENGINEERING_REVIEW.md`).

4.2 Real hardware bugs found & fixed

Issue	Resolution
Invalid pin — <code>SCL = GPIO22</code> does not exist on ESP32-S3	Reassigned I2C to GPIO1/2
USB clash — printer RX on GPIO19 (native USB D+)	Moved printer UART to GPIO17/18
Strapping pin — UV-C enable on GPIO3 (boot strap; unsafe for a safety output)	Moved to GPIO13 + pull-down (off at boot)
I2C clash — two TCS34725 both at address 0x29	Added TCA9548A I2C mux (was missing from BOM)

Unbuildable display — 4.3" MIPI-DSI (ESP32-S3 has no DSI)	Switched to 3.5" SPI TFT (ILI9488)
Power — no 3.3V rail; 9V printer rail referenced but only 5V regulator; ~3.9A peak vs 2.5A regulator	Added AMS1117-3.3; printer at 5V; documented load-sequencing + battery C-rating in POWER_BUDGET.md

4.3 Decisions locked

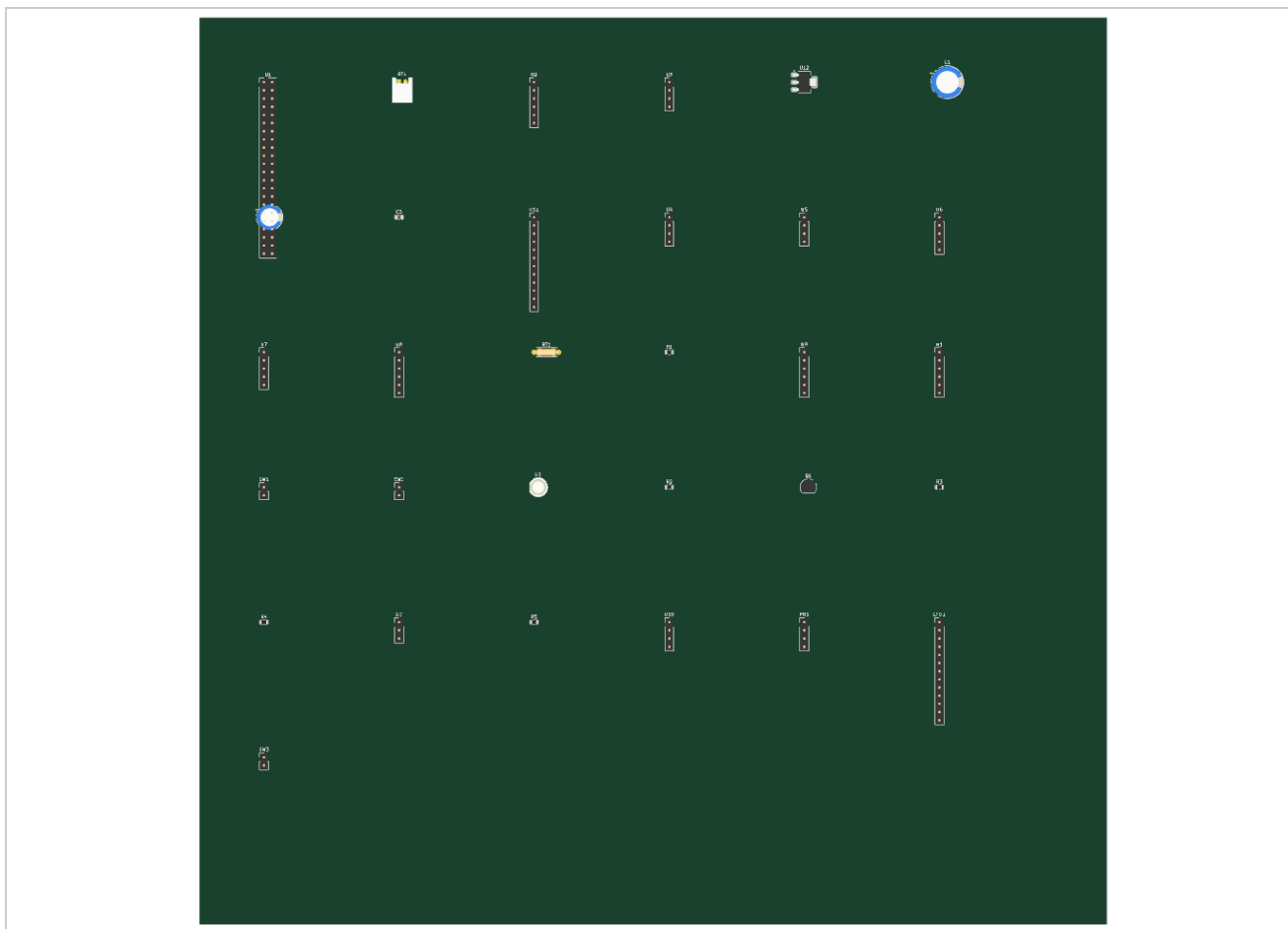
- **Display:** 3.5" SPI TFT + XPT2046 touch.
- **Canonical design:** Design A (reflectance reader). The camera-based KiCad project is a separate, unrelated project.
- **Motor driver:** MX1508. **Printer rail:** 5V.

4.4 Generated & validated artifacts (KiCad 10.0.3)

Artifact	Location	Validation
Netlist (KiCad-importable + table)	hardware/netlist/	31 components, 39 nets
Schematic (real symbols for jellybeans; real footprints)	hardware/schematic/	ERC 0 errors; netlist 39/39 by ref+pad
Starting PCB (placed + net-assigned)	hardware/pcb/	Schematic-parity 0; unrouted ratsnest only

All three are generated from a single source-of-truth model (`gen_netlist.py` + `footprints.py`), so schematic, netlist, and board stay in sync.

4.5 Starting PCB (top render)



31 footprints placed on a grid with nets assigned; intentionally unrouted. Module footprints are 2.54mm pin headers pending datasheet pin-order confirmation.

5. Open decisions & risks

Item	Why it matters	Owner
Proprietary vs commercial strips	Firmware reads standard strips, but stated intent is own strips — a second product (reagent R&D, ISO 23640, separate IVD registration)	Business
Confirm CDSCO class	Class drives which validation tests are owed; wrong class = repeated studies	Reg. consultant
Keep or drop UV-C	275nm is a real eye/skin hazard + IEC 62471 burden; highest novelty-vs-cost trade	Engineering / Risk
Module header pin order	PCB module footprints use netlist order; physical pinouts must be verified vs datasheets before routing	Electronics engineer

6. Next steps

Immediate — electronics engineer (Stage 4 close-out)

1. Review the schematic + `ENGINEERING_REVIEW.md` + `POWER_BUDGET.md`.
2. **Confirm each module's header pin order** against datasheets (`footprints.py`).

3. Swap remaining generic module symbols for preferred library/custom symbols if desired; optionally redraw schematic with wires.
4. Place parts sensibly, add power planes (GND/+5V/+3V3), route (wide traces for motor/printer/UV-C), add chassis mounting holes.
5. **DRC** → **0**; export Gerbers + drill + position files; order 5× prototype PCBs (JLCPCB/PCBWay).

Then — Stage 5 onward

- **Stage 5 Prototype & bench verify:** assemble, flash firmware, calibrate (white standard, scale, pad pitch, hue table), validate scan cycle → **design freeze**.
- **Stage 6 Analytical validation:** LoD, precision, interference, method comparison vs predicate (CLSI EP protocols).
- **Stage 7 Safety/EMC:** IEC 61010, 61326, 62471 (UV-C) at an NABL lab.
- **Stage 8 Human factors + close risk file (ISO 14971/IEC 62366); cybersecurity if Wi-Fi/BT added.**
- **Stage 9 Regulatory (CDSCO):** ISO 13485 QMS, test licence → manufacturing licence; clinical performance evaluation if required.
- **Stages 10–12:** design transfer & manufacturing, pre-launch (labeling/IFU, packaging, pricing), post-market surveillance.

7. Repository map

Path	Contents
regulatory/	Stage 0: TPP, CDSCO classification, standards matrix, ISO 14971 risk file, FTO
research/scispace/	Stage 1: literature + manufacturing/commercial report (2P & 10P, India market)
design/blueprint/	Stage 3: Blueprint mechanical/electrical concept export
hardware/ENGINEERING_REVIEW.md , POWER_BUDGET.md	Stage 4: master pin map + power analysis
hardware/footprints.py	Real KiCad footprints + pad map (single source of truth)
hardware/netlist/	Netlist generator, .net , NETLIST.md
hardware/schematic/	Schematic generator, .kicad_sch , PDF, ERC report
hardware/pcb/	PCB generator, .kicad_pcb , project, render, DRC report
firmware/ , bom/	ESP32-S3 firmware (first draft), India-sourced BOM

Disclaimers. Best-guess CDSCO classification and the FTO read are not legal/regulatory opinions — engage a CDSCO consultant and (before launch/export) a patent attorney. The schematic and PCB are AI-generated first drafts, electrically correct per the model and ERC/parity-validated, but symbol/footprint choices, module pin order, physical layout, and routing require a qualified electronics engineer's sign-off before fabrication.

